

NIZHNEANGARSK, Russian Federation

WMO: 304330

Lat: 55.791N

Lon: 109.581E

Elev: 479

StdP: 95.70

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.8	-29.0	-35.2	0.1	-30.0	-32.9	0.2	-28.1	7.8	-18.6	6.7	-15.8	0.7	10	0.655

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.9	26.4	17.2	24.5	16.5	22.8	15.9	18.7	23.8	17.8	22.4	16.9	21.1	3.0	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.1	12.9	20.5	16.1	12.1	19.7	15.1	11.4	18.9	55.3	24.1	52.4	22.4	49.5	21.1	21.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.8	5.8	5.0	-34.1	29.9	3.6	1.5	-36.7	30.9	-38.8	31.8	-40.8	32.6	-43.4	33.6		
(5)			-34.2	20.4	3.5	0.6	-36.8	20.8	-38.8	21.1	-40.8	21.5	-43.4	21.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.5	-21.0	-19.2	-11.4	-1.6	5.8	13.3	17.7	15.8	8.7	-0.3	-10.5	-17.0
	DBStd	14.19	5.10	5.60	5.67	4.27	3.63	3.98	3.39	3.23	3.71	4.60	6.34	5.55
	HDD10.0	4778	961	817	664	348	139	12	0	2	67	319	615	836
	HDD18.3	7307	1219	1050	922	598	389	159	54	91	288	577	865	1094
	CDD10.0	565	0	0	0	0	9	110	237	181	28	0	0	0
	CDD18.3	52	0	0	0	0	0	7	33	12	0	0	0	0
	CDH23.3	293	0	0	0	0	1	55	189	48	1	0	0	0
	CDH26.7	42	0	0	0	0	0	6	32	4	0	0	0	0
Wind	WSAvg	1.8	1.4	1.2	1.5	2.0	2.1	1.9	1.9	1.9	2.0	1.9	2.0	1.7
Precipitation	PrecAvg	373	16	14	12	20	27	44	60	66	44	33	22	15
	PrecMax	507	64	28	39	51	70	86	155	113	87	83	55	40
	PrecMin	243	3	5	1	3	6	12	14	24	5	0	5	4
	PrecStd	72	12	6	10	12	14	22	37	24	19	20	14	9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-4.9	-4.4	3.9	12.1	20.2	27.5	29.9	26.9	21.8	10.9	3.9	-0.7
		MCWB	-5.7	-6.2	-0.1	4.9	11.0	16.8	18.1	17.4	13.8	6.1	2.1	-2.4
	2%	DB	-8.8	-6.9	1.4	8.7	16.9	24.9	27.3	24.6	18.1	8.4	2.6	-3.9
		MCWB	-9.9	-8.2	-1.7	3.3	8.8	15.6	17.8	16.2	12.4	5.0	1.0	-5.6
	5%	DB	-11.6	-9.0	-0.8	6.5	14.5	22.7	25.4	22.8	16.1	7.0	1.1	-6.3
		MCWB	-12.5	-10.1	-3.4	1.9	7.6	14.9	17.2	15.5	11.4	4.1	-0.8	-7.9
	10%	DB	-13.8	-11.1	-2.9	4.7	12.3	20.3	23.7	21.0	14.3	5.5	-1.2	-8.9
		MCWB	-14.5	-12.0	-4.9	0.8	6.5	14.0	16.8	15.3	10.0	2.8	-3.0	-10.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-5.6	-5.9	0.4	5.9	11.7	18.6	20.1	19.0	15.4	7.6	2.9	-2.0
		MCDB	-4.7	-4.7	3.1	10.8	18.9	24.7	24.4	23.8	18.8	9.3	3.5	-0.7
	2%	WB	-9.9	-8.0	-1.5	3.7	9.7	16.8	19.2	17.6	13.3	5.7	1.2	-5.4
		MCDB	-8.8	-6.9	1.1	7.6	15.4	22.3	24.2	21.7	17.2	7.9	2.4	-3.9
	5%	WB	-12.5	-10.1	-3.3	2.3	8.3	15.7	18.5	16.7	11.8	4.4	-0.8	-7.9
		MCDB	-11.7	-9.2	-1.0	5.7	13.7	21.1	23.4	20.6	15.2	6.5	1.1	-6.4
	10%	WB	-14.4	-11.9	-4.9	1.2	6.9	14.6	17.7	16.0	10.5	3.0	-2.9	-10.1
		MCDB	-13.8	-11.1	-3.0	4.2	11.6	19.3	22.2	19.8	13.6	5.1	-1.3	-8.9
Mean Daily Temperature Range		MDBR	5.2	6.9	9.0	7.8	8.5	8.9	7.9	7.1	6.5	5.5	5.5	5.3
	5% DB	MCDBR	6.7	7.6	10.0	10.0	12.3	12.9	11.0	10.2	9.0	6.9	5.5	7.1
		MCWBR	6.2	6.6	7.5	6.0	6.7	6.1	5.3	5.0	5.0	5.2	5.1	6.3
		MCDBR	6.7	7.6	9.6	9.3	11.2	10.8	9.0	8.1	7.9	6.3	5.7	7.3
	5% WB	MCWBR	6.2	6.7	7.3	5.9	6.5	5.8	5.1	4.7	4.8	5.2	5.6	6.6
Clear-Sky Solar Irradiance	taub	0.244	0.277	0.304	0.344	0.384	0.424	0.419	0.382	0.331	0.289	0.255	0.233	
	taud	2.091	2.084	2.101	2.158	2.179	2.245	2.306	2.390	2.463	2.355	2.120	2.044	
	Ebn at Noon	664	779	853	871	854	821	818	829	832	776	638	571	
	Edh at Noon	60	91	116	128	134	128	118	102	82	70	58	49	
All-Sky Solar Radiation	RadAvg	0.85	1.83	3.44	4.80	5.21	5.67	5.20	4.31	2.86	1.60	0.87	0.60	
	RadStd	0.07	0.12	0.13	0.29	0.41	0.40	0.53	0.33	0.24	0.19	0.10	0.05	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page